

Module 15: Population and Systems Ecology - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Logistic growth differs from exponential growth because logistic growth:
2. A. never slows down
3. B. has no limits of any kind
4. C. levels off as the population nears carrying capacity
5. D. can only ever decrease
6. Answer: C

7. Why: Logistic growth slows as limited resources bring the population toward carrying capacity. Exponential growth assumes unlimited resources.

8. A contagious disease that spreads faster when a population is crowded is an example of a:
9. A. density-dependent factor
10. B. density-independent factor
11. C. trophic level
12. D. decomposer
13. Answer: A

14. Why: Density-dependent factors such as disease and competition grow stronger with crowding. Density-independent factors, like a flood, do not depend on density.

15. About how much of the energy at one trophic level typically passes to the next level up?
16. A. 100 percent
17. B. 90 percent
18. C. 50 percent
19. D. about 10 percent
20. Answer: D

21. Why: Roughly 10 percent of energy transfers to the next level. The rest is lost mostly as heat, which limits how long food chains can be.
22. Energy and matter behave differently in an ecosystem because:
23. A. both are perfectly recycled forever
24. B. matter cycles, but energy flows through and must keep entering
25. C. neither one moves
26. D. energy is recycled but matter is not
27. Answer: B
28. Why: Nutrients cycle within an ecosystem, but energy is lost at each transfer, so it must be continually supplied, for example by sunlight.